



Assignment 1

Assignment must be completed in groups of 2 members and submitted via Google Classroom by 31.08.2026 at 23:59. If the assignment is not prepared using L^AT_EX, a clear scanned copy of the handwritten work must be uploaded. Ensure that the names and enrollment numbers of all group members are clearly written on the submission. Late submissions will not be accepted.

Question 1: [Vector Calculus Operations, 4 Points]

The following vector calculus operations are defined for a scalar function $u : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a vector field $\mathbf{v} = (v_1, v_2, v_3)^\top : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, using the nabla (or del) operator

$$\nabla = \begin{bmatrix} \partial_x & \partial_y & \partial_z \end{bmatrix}^\top.$$

- Gradient:

$$\nabla u = \begin{bmatrix} \partial_x u & \partial_y u & \partial_z u \end{bmatrix}^\top.$$

- Divergence:

$$\nabla \cdot \mathbf{v} = \partial_x v_1 + \partial_y v_2 + \partial_z v_3.$$

- Curl:

$$\nabla \times \mathbf{v} = \begin{bmatrix} \partial_y v_3 - \partial_z v_2 & \partial_z v_1 - \partial_x v_3 & \partial_x v_2 - \partial_y v_1 \end{bmatrix}^\top.$$

- Laplacian:

$$\Delta u = \nabla \cdot (\nabla u) = \partial_{xx} u + \partial_{yy} u + \partial_{zz} u.$$

Which of the following identities are always true, assuming that the functions are sufficiently smooth? Give reasons for your answers.

- a) $\nabla \times (\nabla u) = \mathbf{0}.$
- b) $\nabla \cdot (\nabla \times \mathbf{v}) = 0.$
- c) $\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \Delta \mathbf{v}.$
- d) $\nabla \cdot (u\mathbf{v}) = u\nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla u.$

Question 2: [Classification of PDEs, 4 Points]

Classify the following partial differential equations (PDEs) into elliptic, parabolic, and hyperbolic equations.

- a) $u_{xx} + 2u_{xy} + 2u_{yy} + 4u_{yz} + 5u_{zz} + u_x + u_y = 0.$
-

b) $u_{xx} - e^z u_{xy} + \left(\frac{e^{2z}}{4} + 1\right) u_{yy} - \log(x^2 + y^2 + z^2) u = 0.$

c) $u_{xx} + 4u_{xy} + 3u_{yy} + 3u_x - u_y + 2u = 0.$

d) $a u_{xx} + 2a u_{xy} + a u_{yy} + b u_x + c u_y = 0, \quad a \neq 0.$

Hint: Symmetric positive definite matrices have the property that all eigenvalues are strictly positive.

Question 3: [Dimensionless Stokes Equations, 4 Points]

Consider the steady, incompressible Stokes equations governing creeping flow in a domain Ω :

$$-\mu \Delta \mathbf{u} + \nabla p = \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0,$$

where $\mathbf{u}$ [m s^{-1}] is velocity, p [$\text{Pa} = \text{kg m}^{-1} \text{s}^{-2}$] is pressure, μ [$\text{Pa s} = \text{kg m}^{-1} \text{s}^{-1}$] is dynamic viscosity, and $\mathbf{f}$ [$\text{N m}^{-3} = \text{kg m}^{-2} \text{s}^{-2}$] is a volumetric body force. Note that inertia is absent from these equations, so no scale of the form ρU^2 is available; the only possible balance for the pressure gradient is against the viscous term.

Let L [m], U [m s^{-1}] and F [N m^{-3}] denote characteristic scales for length, velocity and body-force magnitude respectively, and define the dimensionless variables

$$\mathbf{x}' = \frac{\mathbf{x}}{L}, \quad \mathbf{u}' = \frac{\mathbf{u}}{U}, \quad p' = \frac{p}{P_0}, \quad \mathbf{f}' = \frac{\mathbf{f}}{F}.$$

Substitute the scalings into the above equations and write both the equations in fully dimensionless form.

Hint: $\Delta \mathbf{u} = \begin{bmatrix} \Delta u_1 & \Delta u_2 & \Delta u_3 \end{bmatrix}^\top$ if $\Omega \subset \mathbb{R}^3$.

Question 4: [Consistency of Operators, 6 Points]

Let the forward, backward, and central difference operators be denoted by Δ^+ , Δ^- , and $\Delta^\pm$, respectively:

$$\Delta^+ u_i = \frac{u_{i+1} - u_i}{h}, \quad \Delta^- u_i = \frac{u_i - u_{i-1}}{h}, \quad \Delta^\pm u_i = \frac{u_{i+1} - u_{i-1}}{2h},$$

and let the second-order difference operator be denoted by δ^2 :

$$\delta^2 u_i = \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}.$$

Show that

a) $\Delta^\pm u_i = \frac{1}{2}(\Delta^+ u_i + \Delta^- u_i)$ and $\delta^2 u_i = \Delta^+(\Delta^- u_i)$.

b) For a sufficiently smooth function $v(x)$ evaluated at x_i show the estimates

$$\Delta^\pm v_i = v'(x_i) + \mathcal{O}(h^2), \quad \delta^2 v_i = v''(x_i) + \mathcal{O}(h^2).$$

c) Determine the order of consistency of the approximation

$$u''(x) = \frac{1}{12h^2} \left(-u(x+2h) + 16u(x+h) - 30u(x) + 16u(x-h) - u(x-2h) \right).$$

Question 5: [Ordering of Nodes, 6 Points]

Consider the two-dimensional Poisson equation

$$-\Delta u = f$$

discretized on a square domain using the standard 5-point finite difference stencil with zero Dirichlet boundary conditions. Let the domain be divided into $M = 4$ subintervals in each direction, yielding a 3×3 grid of interior nodes (i, j) for $i, j \in \{1, 2, 3\}$.

Suppose that, instead of the natural column-by-column ordering, an odd-even (red-black) ordering is used: unknown grid values $u_{i,j}$ with $i + j$ even are ordered column-wise into a vector $\mathbf{u}_e$, while those with $i + j$ odd form a vector $\mathbf{u}_o$.

- (a) List the grid points belonging to $\mathbf{u}_e$ and $\mathbf{u}_o$ in the prescribed ordering.
- (b) Write down the resulting coefficient matrix $\mathbf{A}$ in the form

$$\mathbf{A} = \begin{bmatrix} 4\mathbb{I}_e & \mathbf{B} \\ \mathbf{B}^\top & 4\mathbb{I}_o \end{bmatrix},$$

and explicitly give the matrix $\mathbf{B}$. Here $\mathbb{I}_e$ and $\mathbb{I}_o$ denote the identity matrix corresponding to the $\mathbf{u}_e$ and $\mathbf{u}_o$, respectively,

- (c) Briefly explain why there are no nonzero off-diagonal entries within either the even or odd blocks.

Question 6: [Non-Constant Diffusion, 6 Points]

Consider the differential operator

$$\mathcal{L}u = \frac{\partial}{\partial x} \left(k(x) \frac{\partial u}{\partial x} \right), \quad (1)$$

where $k \in \mathcal{C}^4$ and the finite-difference approximation

$$(\mathcal{L}_h \mathbf{u})_i = \frac{1}{h} \left(a_{i+1} \frac{\mathbf{u}_{i+1} - \mathbf{u}_i}{h} - a_i \frac{\mathbf{u}_i - \mathbf{u}_{i-1}}{h} \right), \quad (2)$$

on a uniform grid $x_i = x_0 + ih$ (assume u is sufficiently smooth). Here a_i is a grid function that approximates k at appropriate locations.

Show that the choices

a) $a_i = \frac{k_i + k_{i-1}}{2}$, and

b) $a_i = k \left(x_i - \frac{h}{2} \right)$,

each yield second-order consistency of $\mathcal{L}_h$ with $\mathcal{L}$ (i.e. the local truncation error is $\mathcal{O}(h^2)$).

Hints:

1. Use chain rule to write $\mathcal{L}u$ in terms of u_x and u_{xx} .

2. Use forward (Δ^+) and backward (Δ^-) difference approximation to write $(\mathcal{L}_h u)_i$ in terms of $u_x|_i$, $u_{xx}|_i$, and $u_{xxx}|_i$.
3. The local truncation error is given by $\mathcal{L}u|_i - \mathcal{L}_h u|_i$.
4. From point 3. you will get the consistency constraints

$$\frac{a_{i+1} - a_i}{h} = k'(x_i) + \mathcal{O}(h^2), \quad \frac{a_{i+1} + a_i}{2} = k(x_i) + \mathcal{O}(h^2).$$

5. Substitute the value of a_i to get the answer. You might need to expand k_{i+1} and k_{i-1} .

Question 7: [Programming, 10 Points]

Consider the Poisson equation

$$\begin{aligned} -\Delta u &= f & \text{in } \Omega, \\ u &= g & \text{on } \Gamma, \end{aligned}$$

where $\Gamma = \partial\Omega$ denotes the boundary of the domain.

We have derived the five-point stencil for solving the above PDE in the case $\Omega = [0, 1]^2$ with uniform grid spacing. In the attached Python script, you will find the code for this case.

The right-hand side f and the boundary condition g are chosen such that the exact solution is

$$u(x, y) = x^4 y^5 - 17 \sin(xy).$$

From theory, we expect a convergence rate of order 2, which can be verified numerically in the provided code.

For bonus points, extend the implementation as follows:

- a) Modify the code so that the five-point stencil can be applied on a rectangular domain, e.g. $\Omega = [0, 1] \times [0, 2]$, using uniform grid spacing.
- b) Modify the code to allow for non-uniform grid spacing with mesh sizes h_x and h_y in the x - and y -directions, respectively.
